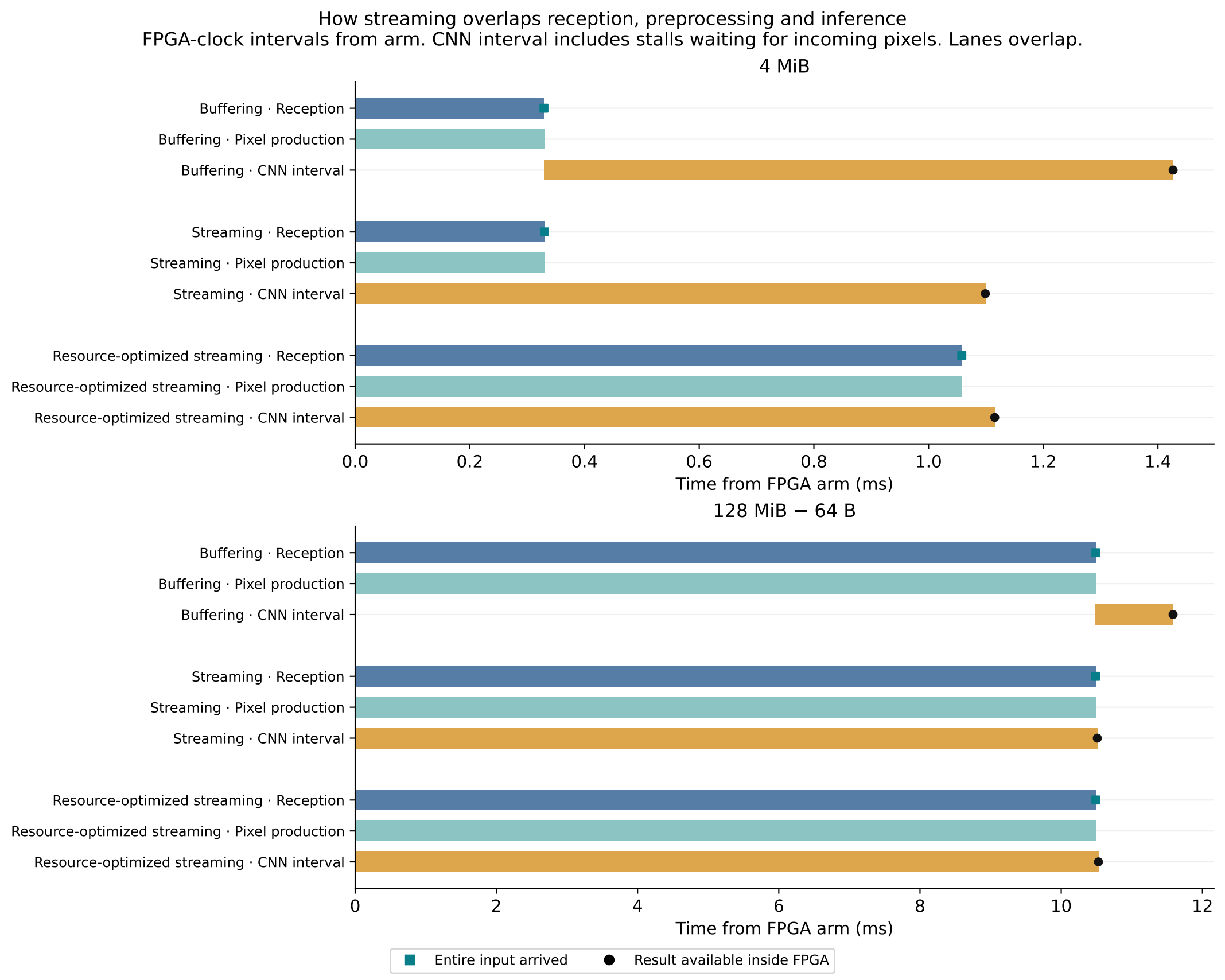
How streaming overlaps reception, preprocessing and inference
FPGA-clock intervals from arm. CNN interval includes stalls waiting for incoming pixels. Lanes overlap.
4 MiB
Buffering · Reception
Buffering · Pixel production
Buffering · CNN interval
Streaming · Reception
Streaming · Pixel production
Streaming · CNN interval
Resource-optimized streaming · Reception
Resource-optimized streaming · Pixel production
Resource-optimized streaming · CNN interval
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Time from FPGA arm (ms)
128 MiB − 64 B
Buffering · Reception
Buffering · Pixel production
Buffering · CNN interval
Streaming · Reception
Streaming · Pixel production
Streaming · CNN interval
Resource-optimized streaming · Reception
Resource-optimized streaming · Pixel production
Resource-optimized streaming · CNN interval
0
2
4
6
8
10
12
Time from FPGA arm (ms)
Entire input arrived
Result available inside FPGA